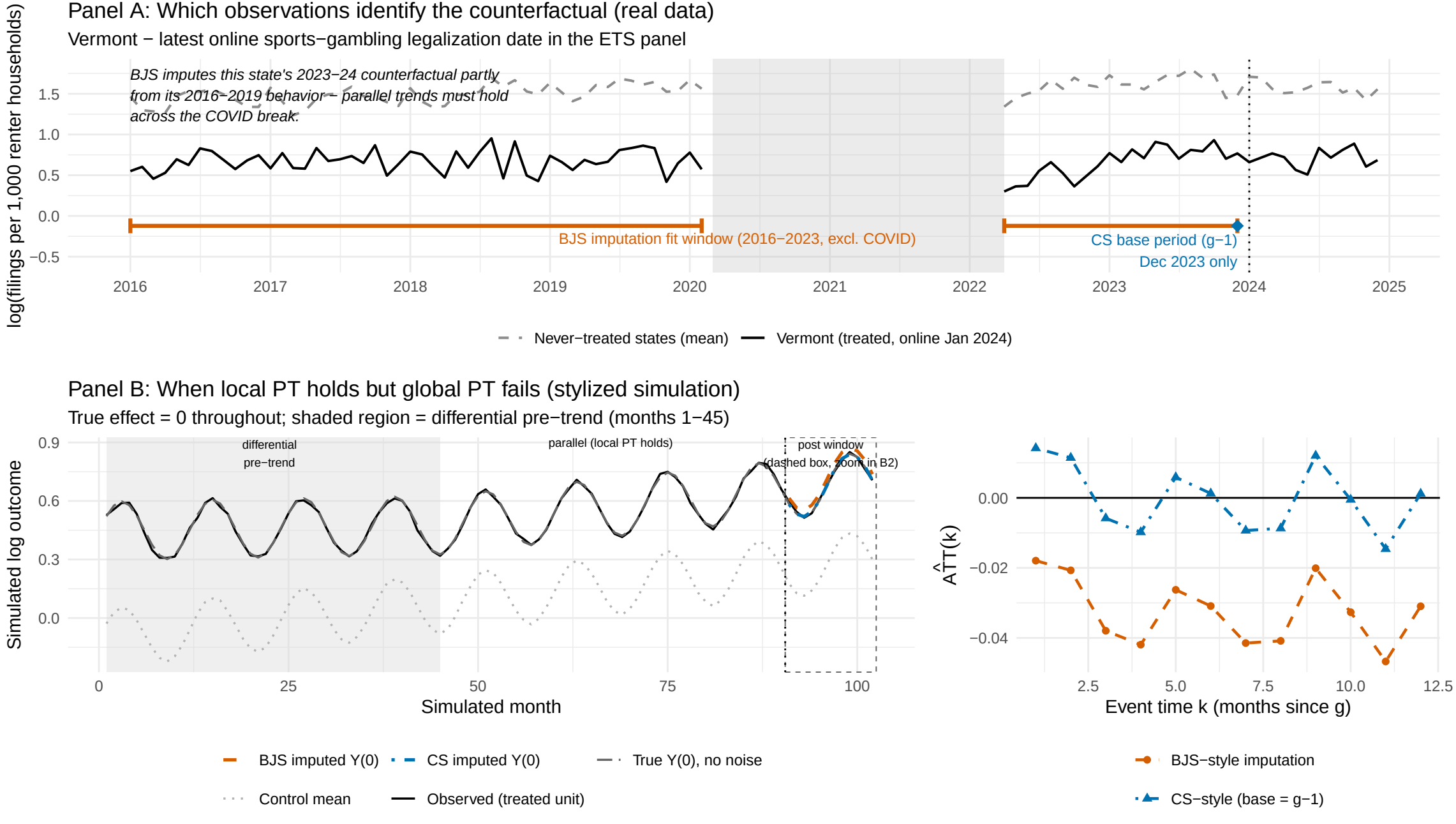
Panel A: Which observations identify the counterfactual (real data)
Vermont − latest online sports−gambling legalization date in the ETS panel
log(filings per 1,000 renter households)
BJS imputes this state's 2023−24 counterfactual partly from its 2016−2019 behavior − parallel trends must hold across the COVID break.
BJS imputation fit window (2016−2023, excl. COVID)
CS base period (g−1)
Dec 2023 only
Never−treated states (mean)
Vermont (treated, online Jan 2024)
Panel B: When local PT holds but global PT fails (stylized simulation)
True effect = 0 throughout; shaded region = differential pre−trend (months 1−45)
Simulated log outcome
differential pre−trend
parallel (local PT holds)
post window (dashed box, zoom in B2)
Simulated month
BJS imputed Y(0)
CS imputed Y(0)
True Y(0), no noise
Control mean
Observed (treated unit)
ATT(k)
Event time k (months since g)
BJS−style imputation
CS−style (base = g−1)